

Generative AI Models for Functional Data

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Contents

- 1 Introduction and Objectives
- 2 Related Work
- 3 Diffusion Models
- 4 Flow Matching
- 5 Empirical Evaluation
- 6 Conclusions and Future Work
- 7 References

Introduction

Question

Why generative AI models?

- Synthetic data generation.
- Data scarcity: expensive and limited.



Figure: Image
(Source: OpenAI)

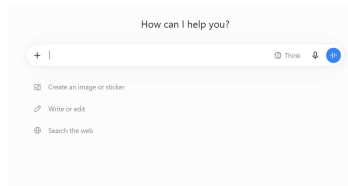


Figure: Text
(Source: ChatGPT)

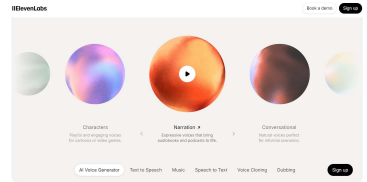


Figure: Audio
(Source: ElevenLabs)

Why functional data?

- Most real-world scenarios.
- Functions of a continuous argument.



Framework

**Functional Data Analysis
(FDA)**

Research Question

Can generative AI models be adapted to functional data?

Objectives

Research Question

Can generative AI models be adapted to functional data?

Proposed Approach

Generate realistic and diverse functional data using modern generative frameworks:

- Flow Matching.
- Diffusion Models.

Evaluation Strategy

Experimental validation on both synthetic and real-world datasets.

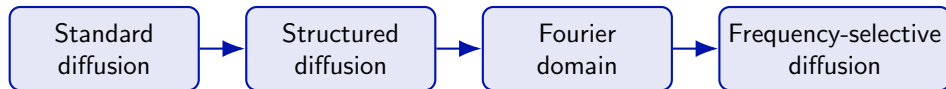
Objectives: Focus on Diffusion Models

Standard Diffusion

Adapt standard diffusion models to functional data.

Structured Diffusion

Introduce structured noise injection to control diffusion across frequencies.



Related Work: Functional Data Analysis (FDA)

Mathematical Formulation

FDA is the branch of statistics for data represented by **functions** of a **continuous parameter**:

$$x : \mathcal{T} \subset \mathbb{R}^p \longrightarrow \mathbb{R}^q, \quad \tau \longrightarrow x(\tau).$$

Observed on a finite grid $\mathbf{t} = (t_1, \dots, t_M) \in \mathcal{T}^M$:

$$x(\mathbf{t}) = (x(t_1), \dots, x(t_M)).$$

Related Applications

- Time series forecasting [1].
- Electroencephalograms [2].

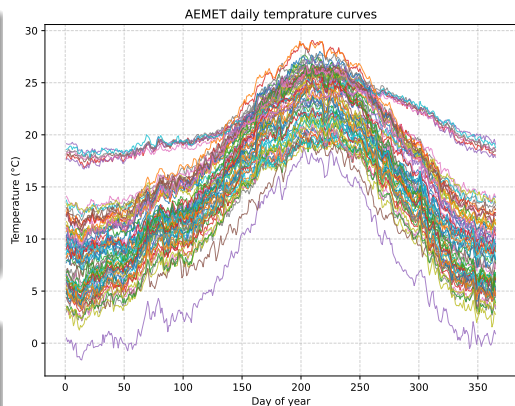


Figure: AEMET temperature curves [3]

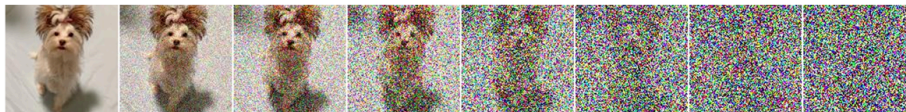
Related Work: Score-Based Diffusion Models

Related Papers

- Song & Ermon (2019) [4]: score-based generative modelling.
- Ho et al. (2020) [5]: DDPM.
- Song et al. (2021) [6]: continuous-time SDE framework.

Forward SDE: data $\rightarrow$ noise

$$\mathbf{x}(0) \rightarrow d\mathbf{x}(\tau) = f(\mathbf{x}(\tau), \tau)d\tau + g(\tau)d\mathbf{W}(\tau) \rightarrow \mathbf{x}(T)$$



Backward SDE: data $\leftarrow$ noise

$$\mathbf{x}(0) \leftarrow d\bar{\mathbf{x}}(\tau) = (f(\bar{\mathbf{x}}(\tau), \tau) - g^2(\tau)\nabla_{\bar{\mathbf{x}}} \log p_{\tau}(\bar{\mathbf{x}}))d\tau + g(\tau)d\bar{\mathbf{W}}(\tau) \leftarrow \mathbf{x}(T)$$

Figure: Forward and reverse processes in score-based diffusion models (Song et al., 2021) [6]

Standard diffusion ($\Sigma \propto I_D$) $\longrightarrow$ **Structured diffusion**

Functional data exhibit structural properties such as **continuity and smoothness**.

Structured Diffusion Approaches

- **Physical constraints** (e.g., heat dissipation/blurring) [7, 8].
- **Adaptive & Learnt structure** (e.g., spatial or learnt covariance) [9, 10].
- **Spectral domain** (e.g., frequency-diagonal covariance) [11, 12].

Related Work: Flow Matching

Diffusion Models

- **Process:** Stochastic (SDE).
- **Forward:** Data $\rightarrow$ Noise.
- **Generation:** Noise $\rightarrow$ Data.
- **Objective:** Learn the **score**.

Flow Matching

- **Process:** Deterministic (ODE).
- **Transport:** Noise $\rightarrow$ Data.
- **Objective:** Learn the **velocity field**.



$\hookrightarrow$ We apply Flow Matching to functional data, to evaluate its performance and compare it with diffusion models.

Diffusion Models: Variance Preserving (VP) Model

Noise Injection

Forward SDE:

$$d\mathbf{x}(\tau) = -\lambda(\tau) \mathbf{x}(\tau) d\tau + \sigma(\tau) d\mathbf{W}(\tau), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \sim p_0(\mathbf{x}), \quad \tau \in [t_0, T].$$

Solution:

$$\mathbf{x}(\tau) \sim \mathcal{N}_D \left(e^{-\int_{t_0}^{\tau} \lambda(u) du} \mathbf{x}_0, \left(\int_{t_0}^{\tau} e^{-2 \int_u^{\tau} \lambda(r) dr} \sigma(u)^2 du \right) I_D \right).$$

Asymptotic Gaussian Distribution

Constraint: $\sigma(\tau)^2 = 2 \lambda(\tau) \implies \mathbf{x}(\tau) \xrightarrow{\tau \rightarrow \infty} \mathcal{N}_D(\mathbf{0}, I_D).$

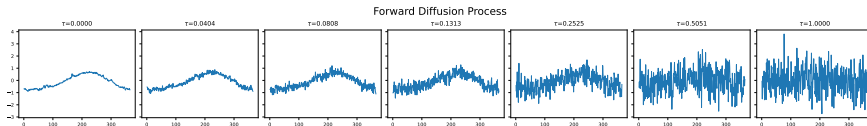


Figure: Forward diffusion process applied to AEMET data

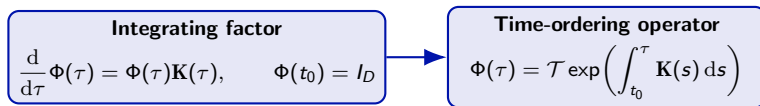
Diffusion Models: From Isotropic to Structured Diffusion

Idea: $\lambda(\tau) I_D \longrightarrow \mathbf{K}(\tau), \quad \sigma(\tau) I_D \longrightarrow \mathbf{S}(\tau).$

Noise Injection

Forward multivariate SDE:

$$d\mathbf{x}(\tau) = -\mathbf{K}(\tau)\mathbf{x}(\tau) d\tau + \mathbf{S}(\tau) d\mathbf{W}(\tau), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \sim p_0(\mathbf{x}), \quad \tau \in [t_0, T].$$



Asymptotic Gaussian Distribution

$$\Sigma_{\infty} = I_D \quad \Longleftrightarrow \quad \mathbf{K}_{\infty} + \mathbf{K}_{\infty}^{\top} = \mathbf{S}_{\infty} \mathbf{S}_{\infty}^{\top}.$$

Diffusion Models: Circulant Diffusion

Fourier Domain

1. **Diagonalisation:** $\mathbf{K}(\tau) = \mathbf{F}^\dagger \mathbf{\Lambda}(\tau) \mathbf{F}$, $\mathbf{S}(\tau) = \mathbf{F}^\dagger \mathbf{B}(\tau) \mathbf{F}$.

2. **Decoupling in Fourier space:**

$$\mathbf{y}(\tau) = \mathbf{F} \mathbf{x}(\tau), \quad \widetilde{\mathbf{W}}(\tau) = \mathbf{F} \mathbf{W}(\tau) \implies \boxed{\mathrm{d}\mathbf{y}(\tau) = -\mathbf{\Lambda}(\tau) \mathbf{y}(\tau) \mathrm{d}\tau + \mathbf{B}(\tau) \mathrm{d}\widetilde{\mathbf{W}}(\tau)}$$

Since $\mathbf{\Lambda}(\tau)$ and $\mathbf{B}(\tau)$ are diagonal, the system is decoupled into D scalar SDEs.

3. **Solution:**

$$\boxed{\mathbf{x}(\tau) \sim \mathcal{N}\left(e^{-\int_{t_0}^{\tau} \mathbf{K}(u) \mathrm{d}u} \mathbf{x}_0, \int_{t_0}^{\tau} e^{-\int_u^{\tau} (\mathbf{K}(r) + \mathbf{K}(r)^\top) \mathrm{d}r} \mathbf{S}(u) \mathbf{S}(u)^\top \mathrm{d}u\right)}$$

$$\lambda_m(\tau) = \lambda_{-m}(\tau) \geq 0, \\ \beta_m(\tau)^2 = 2\lambda_m(\tau)$$

$$\mathbf{K}(\tau), \mathbf{S}(\tau) \text{ symmetric,} \\ \mathbf{x}(\tau) \xrightarrow{\tau \rightarrow \infty} \mathcal{N}_D(\mathbf{0}, I_D)$$

Standard Diffusion

- **Backward SDE:**

$$d\overleftarrow{\mathbf{x}}(\tau) = (f(\overleftarrow{\mathbf{x}}(\tau), \tau) - g^2(\tau) \nabla_{\overleftarrow{\mathbf{x}}} \log p_\tau(\overleftarrow{\mathbf{x}})) d\tau + g(\tau) d\overleftarrow{\mathbf{W}}(\tau), \quad \mathbf{x}(T) \sim p_1(\mathbf{x}), \quad \tau \in [T, t_0].$$

- **Conditional score:**

$$\mathbf{x}(\tau) = \boldsymbol{\mu}_{\mathbf{x}_0}(\tau) + \sigma(\tau) \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}_D(\mathbf{0}, I_D) \implies \nabla_{\mathbf{x}} \log p(\mathbf{x}, \tau \mid \mathbf{x}_0, 0) = -\sigma(\tau)^{-1} \mathbf{z}.$$

- **Loss function:**

$$\mathcal{L}(\theta) \propto \mathbb{E} [\|\sigma(\tau) \mathbf{s}(\mathbf{x}, \tau; \theta) + \mathbf{z}\|_2^2].$$

- **Sample generation:**

$$d\overleftarrow{\mathbf{x}}(\tau) = (f(\overleftarrow{\mathbf{x}}(\tau), \tau) - g^2(\tau) \mathbf{s}_\theta(\overleftarrow{\mathbf{x}}(\tau), \tau)) d\tau + g(\tau) d\overleftarrow{\mathbf{W}}(\tau), \quad \mathbf{x}(T) \sim p_1(\mathbf{x}), \quad \tau \in [T, t_0].$$

Structured Diffusion

- **Backward SDE:**

$$d\overleftarrow{\mathbf{x}}(\tau) = \left(\mathbf{f}(\overleftarrow{\mathbf{x}}(\tau), \tau) - \mathbf{G}(\tau)\mathbf{G}(\tau)^\top \nabla_{\overleftarrow{\mathbf{x}}} \log p_\tau(\overleftarrow{\mathbf{x}}) \right) d\tau + \mathbf{G}(\tau) d\overleftarrow{\mathbf{W}}(\tau), \quad \mathbf{x}(T) \sim p_1(\mathbf{x}), \tau \in [T, t_0].$$

- **Conditional score:**

$$\mathbf{x}(\tau) = \boldsymbol{\mu}_{\mathbf{x}_0}(\tau) + \mathbf{L}_\tau \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}_D(\mathbf{0}, I_D) \implies \nabla_{\mathbf{x}} \log p(\mathbf{x}, \tau \mid \mathbf{x}_0, 0) = -(\mathbf{L}_\tau^\top)^{-1} \mathbf{z}.$$

- **Loss function:**

$$\mathcal{L}(\theta) \propto \mathbb{E} \left[\|\mathbf{L}_\tau^\top \mathbf{s}(\mathbf{x}, \tau; \theta) + \mathbf{z}\|_2^2 \right].$$

- **Sample generation:**

$$d\overleftarrow{\mathbf{x}}(\tau) = \left(\mathbf{f}(\overleftarrow{\mathbf{x}}(\tau), \tau) - \mathbf{G}(\tau)\mathbf{G}(\tau)^\top \mathbf{s}_\theta(\overleftarrow{\mathbf{x}}(\tau), \tau) \right) d\tau + \mathbf{G}(\tau) d\overleftarrow{\mathbf{W}}(\tau), \quad \mathbf{x}(T) \sim p_1(\mathbf{x}), \tau \in [T, t_0].$$

Flow Matching: From Diffusion to Flow Matching

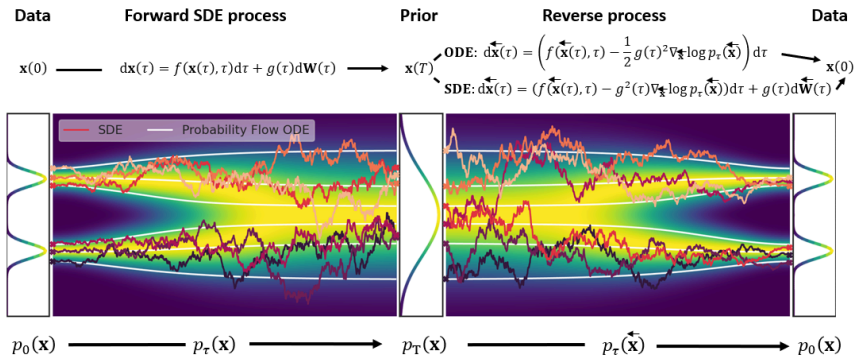


Figure: Probability flow ODE (Song et al., 2021) [6]

Question

Is it possible to build a deterministic transport between noise and the original data?

Flow Matching

Idea

$$\text{Noise} \sim p_0(\mathbf{x}) \xrightarrow{\text{Continuous flow via ODE}} \text{Data} \sim p_1(\mathbf{x})$$
$$\frac{d\mathbf{x}(\tau)}{d\tau} = \mathbf{v}(\mathbf{x}(\tau), \tau), \quad \tau \in [0, 1].$$

Conditional Flow Matching

Define a target trajectory between $\mathbf{x}_0 \sim p_0(\mathbf{x})$ and $\mathbf{x}_1 \sim p_1(\mathbf{x})$:

$$\frac{d\mathbf{x}(\tau)}{d\tau} = \mathbf{v}_c(\mathbf{x}(\tau), \tau \mid \mathbf{x}_0, \mathbf{x}_1), \quad \mathbf{x}(0) = \mathbf{x}_0, \quad \mathbf{x}(1) = \mathbf{x}_1.$$

Training

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{v}_\theta(\mathbf{x}(\tau), \tau) - \mathbf{v}_c(\mathbf{x}(\tau), \tau \mid \mathbf{x}_0, \mathbf{x}_1)\|^2 \right].$$

Generation

$$\frac{d\mathbf{x}(\tau)}{d\tau} = \mathbf{v}_\theta(\mathbf{x}(\tau), \tau), \quad \mathbf{x}(0) \sim p_0(\mathbf{x}).$$

Flow Matching: Linear Flow Matching

Linear Interpolation Scheme

For $\tau \in [0, 1]$, $\mathbf{x}_0 \sim p_0(\mathbf{x})$, $\mathbf{x}_1 \sim p_1(\mathbf{x})$:

$$\mathbf{x}(\tau) = (1 - \tau)\mathbf{x}_0 + \tau\mathbf{x}_1 \implies \mathbf{v}_c(\mathbf{x}(\tau), \tau \mid \mathbf{x}_0, \mathbf{x}_1) = \mathbf{x}_1 - \mathbf{x}_0$$

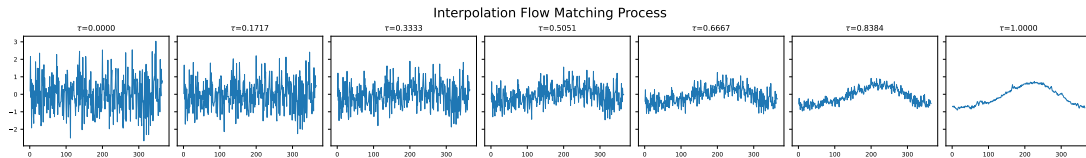


Figure: Linear interpolation path from noise ($\tau = 0$) to AEMET data ($\tau = 1$).

Training

$$\mathcal{L}(\theta) = \mathbb{E}_{\tau, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{v}_\theta(\mathbf{x}(\tau), \tau) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2 \right].$$

Generation

$$\frac{d\mathbf{x}(\tau)}{d\tau} = \mathbf{v}_\theta(\mathbf{x}(\tau), \tau), \quad \mathbf{x}(0) \sim p_0(\mathbf{x}).$$

Experimental Framework

- 6 experiments: AEMET $\rightarrow$ synthetic controlled data $\rightarrow$ ECG.

Generative Models

- Standard diffusion: $dx(\tau) = -\lambda(\tau) x(\tau) d\tau + \sigma(\tau) d\mathbf{W}(\tau)$, $\mathbf{x}(t_0) = \mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $\tau \in [0, 1]$.
- Flow Matching: $\mathbf{x}(\tau) = (1 - \tau) \mathbf{x}_0 + \tau \mathbf{x}_1$, $\mathbf{x}_0 \sim \mathcal{N}_D(\mathbf{0}, I_D)$, $\mathbf{x}_1 \sim p_{\text{data}}(\mathbf{x})$, $\tau \in [0, 1]$.
- Circulant diffusion model (experiment 3):

$$dx(\tau) = -\mathbf{K}(\tau) \mathbf{x}(\tau) d\tau + \mathbf{S}(\tau) d\mathbf{W}(\tau), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}), \quad \tau \in [0, 1].$$

- Diffusion noise injection: $\lambda(\tau) = \lambda_{\min} + \tau(\lambda_{\max} - \lambda_{\min})$.
- Architecture: 1D U-Net with time-embedding.
- Hyperparameters selected experimentally.

Preliminary Experiment: AEMET

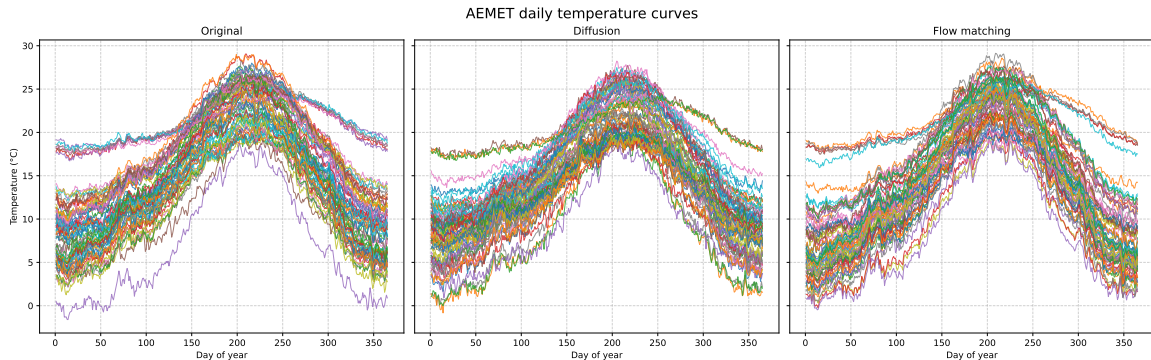


Figure: Real and generated AEMET curves

Experiment 1: Constant Functions

Dataset

$$\mathbf{x}(t) = c, \quad t \in [0, 2\pi], \quad c \sim U[-1, 1].$$

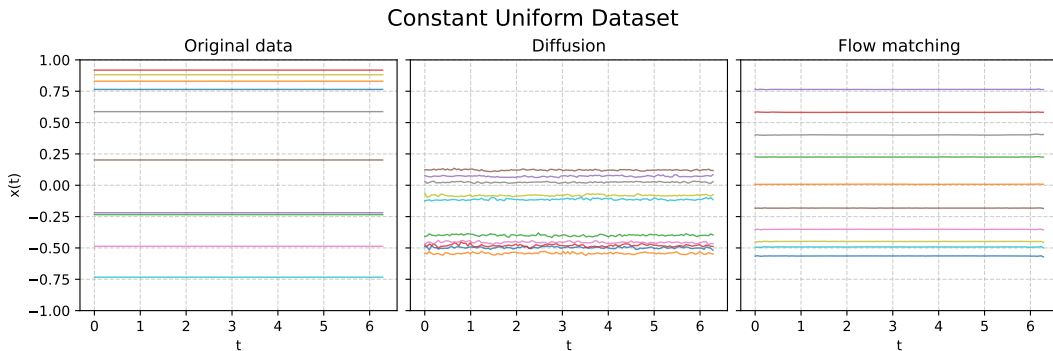


Figure: Real and generated constant functions

Experiment 1: Constant Functions

Quality Metrics

- MSE: $\text{MSE}_i = \frac{1}{M} \sum_{m=1}^M (\hat{x}_i(t_m) - \hat{c}_i)^2$, $\hat{c}_i = \frac{1}{M} \sum_{m=1}^M \hat{x}_i(t_m)$.
- Wasserstein distance: $W_1(\mathcal{C}, \mathcal{U}(-1, 1)) = \int_{\mathbb{R}} |F_{\mathcal{C}}(x) - F_{\mathcal{U}}(x)| dx$.

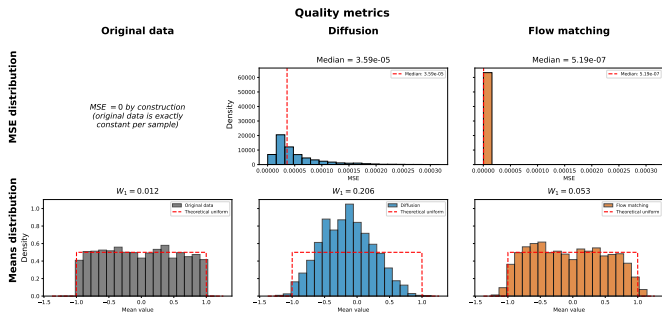


Figure: Quality metrics for constant functions experiment

Experiment 2: Damped Oscillatory Functions

Dataset

$$\mathbf{x}(t) = a + b e^{-\alpha t} \cos(\omega t + \phi), \quad t \in [0, 2\pi],$$
$$a \sim \mathcal{U}(-1, 1), \quad b \sim \mathcal{U}(-1, 1), \quad \alpha \sim \mathcal{U}(0.1, 1), \quad \omega \sim \mathcal{U}(2, 4), \quad \phi \sim \mathcal{U}\left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Exponential Decay Cosine Dataset

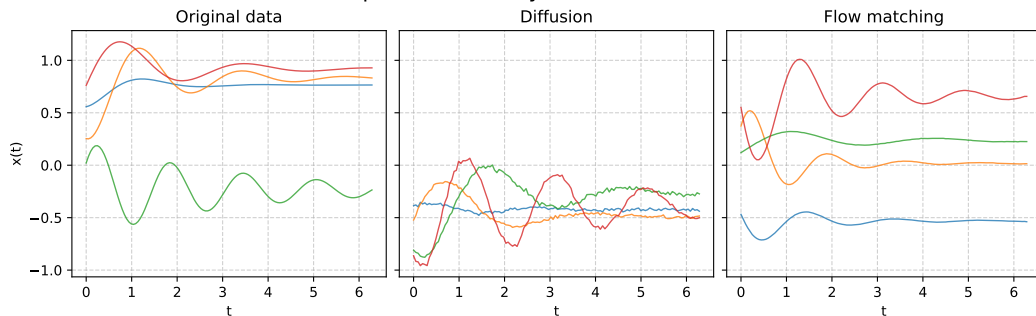


Figure: Real and generated damped oscillatory functions

Experiment 2: Damped Oscillatory Functions

Evaluation Setup

- **Parametric model:** $f(t, \theta) = a + b e^{-\alpha t} \cos(\omega t + \phi)$, $\theta = (a, b, \alpha, \omega, \phi)$.
- **Quality metrics:** MSE and Wasserstein distance.

Model	MSE (median)
Diffusion	1.60×10^{-4}
Flow Matching	6.84×10^{-6}

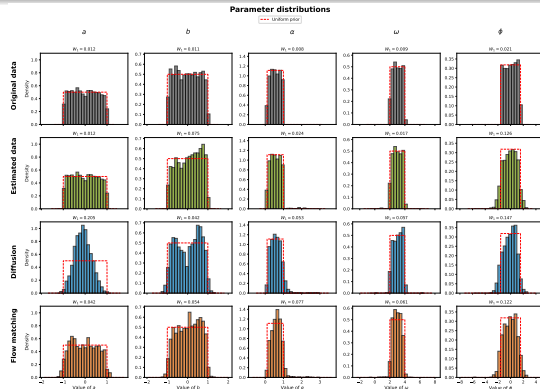


Figure: Wasserstein distance of parameters distributions

Experiment 3: Frequency-Selective Diffusion

Dataset

$x(t) = \cos(2\pi t + \phi) + 0.3 \cos(14\pi t + \phi)$
with $t \in [0, 1]$ and $\phi \sim \mathcal{U}(0, 2\pi)$.

- **Low freq:** $\omega_{\text{low}} = 1$, $A_{2\pi} = 1.0$.
- **High freq:** $\omega_{\text{high}} = 7$, $A_{14\pi} = 0.3$.

Structured Circulant Diffusion

- **Low-frequency mode:** low frequencies decay first, high later.
- **High-frequency mode:** high frequencies decay first, low later.

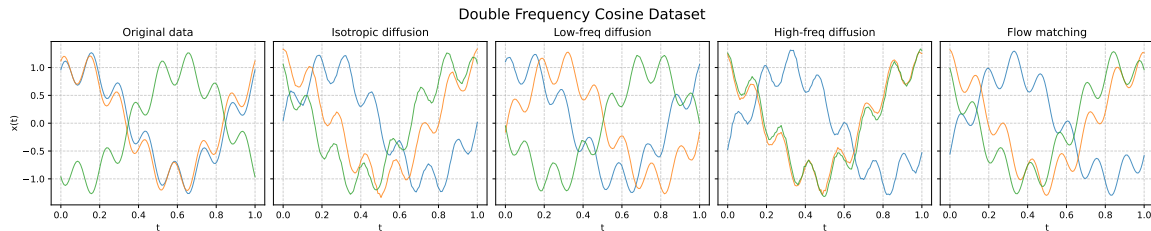


Figure: Real and generated double-frequency cosine functions

Experiment 3: Frequency-Selective Diffusion

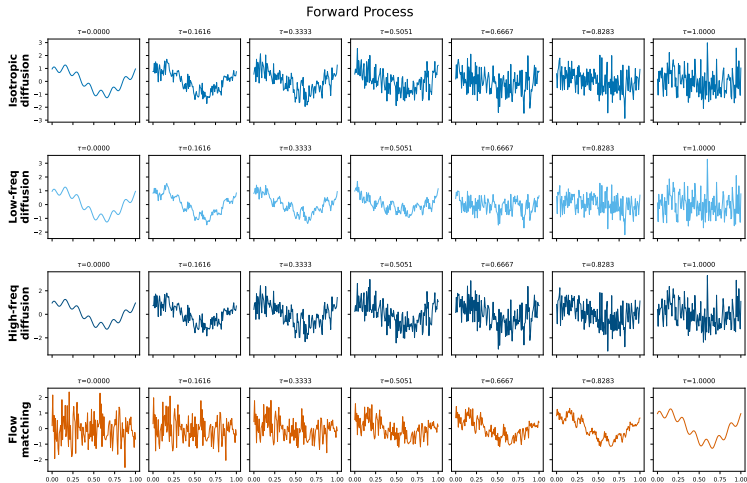


Figure: Forward process for diffusion and Flow Matching models

Experiment 3: Frequency-Selective Diffusion

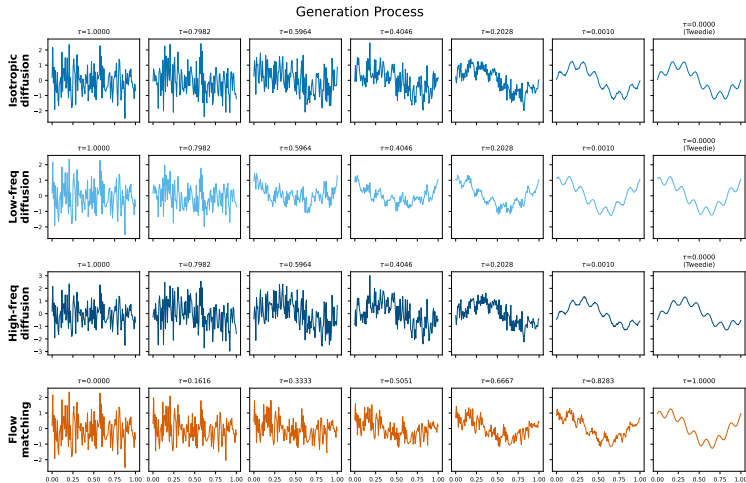


Figure: Generation process for diffusion and Flow Matching models

Experiment 3: Frequency-Selective Diffusion

Evaluation Setup

- **Phase model:** $f(t, \phi) = \cos(2\pi t + \phi) + 0.3 \cos(14\pi t + \phi)$, with $\phi \sim \mathcal{U}(0, 2\pi)$.
- **Amplitude model:** $g_i(t, a_{2\pi}, a_{14\pi}) = a_{2\pi} \cos(2\pi t + \phi_i^*) + a_{14\pi} \cos(14\pi t + \phi_i^*)$, $a_{2\pi}, a_{14\pi} \geq 0$.
- **Quality metrics:** MSE, phase Wasserstein distance ($W_1(\phi)$), recovered amplitudes ($a_{2\pi}, a_{14\pi}$).

Table: Quality metrics for double-frequency cosine experiment

Model	MSE (median)	$W_1(\phi)$	$a_{2\pi}$ (mean $\pm$ std)	$a_{14\pi}$ (mean $\pm$ std)
Original data	-	0.037	1.000 ± 0.000	0.300 ± 0.000
Estimated data	-	0.037	1.000 ± 0.000	0.300 ± 0.000
Isotropic diffusion	3.66e-04	0.077	1.010 ± 0.009	0.300 ± 0.005
Low-frequency diffusion	2.70e-04	0.179	1.009 ± 0.010	0.298 ± 0.004
High-frequency diffusion	4.25e-04	0.077	1.005 ± 0.007	0.300 ± 0.006
Flow matching	1.15e-04	0.052	1.007 ± 0.010	0.301 ± 0.004

Experiment 4: Skyline Functions

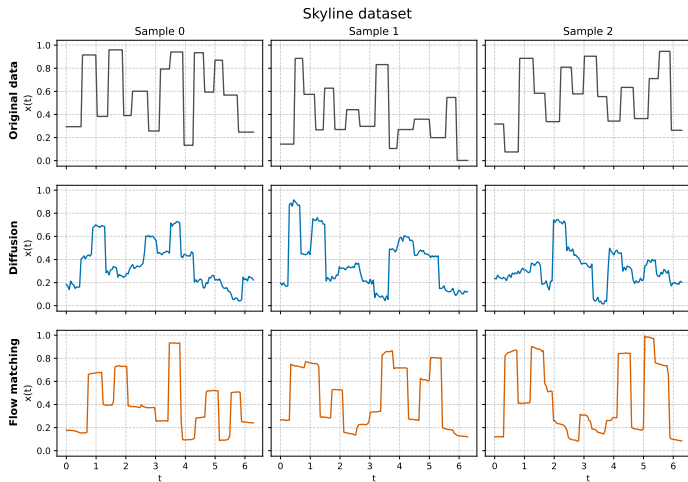


Figure: Real and generated skyline functions

Experiment 5: ECG Signals

Dataset

- **PTB-XL ECG database** [15].
- Normal ECG recordings only.
- $N = 9246$ signals, $M = 500$ points (5 s, 100 Hz).

Quality Metrics

What should a realistic ECG reproduce?

- Detection success rate.
- RR intervals (s): μ_{RR}, σ_{RR} .
- Heart rate (bpm): $HR = \frac{60}{\mu_{RR}}$.

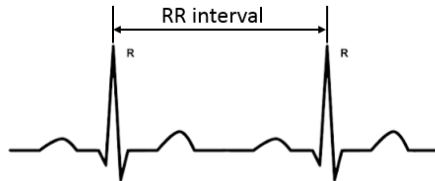


Figure: Main components of an ECG signal

Experiment 5: ECG Signals

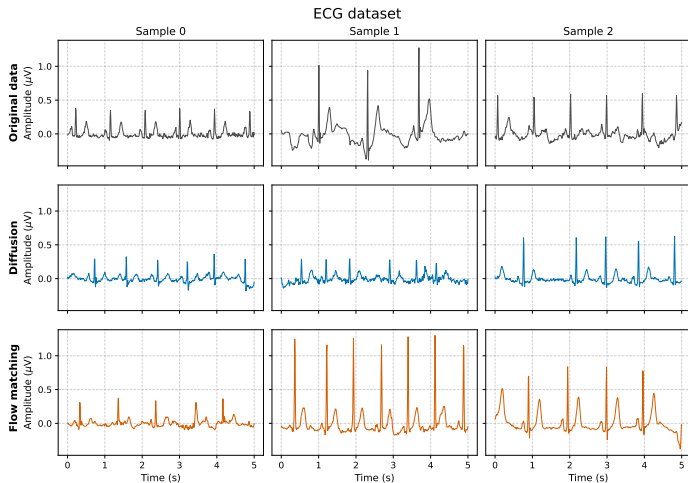


Figure: Real and generated ECG signals

Experiment 5: ECG Signals

Table: Quality metrics for ECG signals experiment

Model	Success (%)	Heart rate Mean $\pm$ Std	Mean RR interval Mean $\pm$ Std	Std RR interval Mean $\pm$ Std
Original data	93.1	96.14 $\pm$ 28.54	0.68 $\pm$ 0.20	0.11 $\pm$ 0.08
Diffusion	90.5	92.61 $\pm$ 26.19	0.70 $\pm$ 0.19	0.15 $\pm$ 0.08
Flow Matching	93.7	86.88 $\pm$ 23.55	0.74 $\pm$ 0.19	0.16 $\pm$ 0.09

Conclusions

Main Question

Can generative AI models be adapted to functional data?

Conclusions

The conclusions achieved in this work are:

- Generative modelling is feasible for functional data.
- Flow Matching generally outperforms diffusion models in our experiments.
- Circulant diffusion enables spectral decoupling and frequency-selective control.
- Methodological insight: added complexity allows for interpretability and control, but not necessarily better results.

Immediate Extensions

- Train the diffusion model directly in the Fourier domain.
- Alternative noise injection schedules for diffusion models.

Further Research

- Latent diffusion for functional data.
- Advanced Flow Matching.

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Thank you for your attention!

Questions are welcome.